

Locality for Codes over the Integers

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Codes over the Integers

Codes over the Integers



Integer codes: $\mathcal{C} \subseteq \mathbb{Z}/q_1\mathbb{Z} \times \cdots \times \mathbb{Z}/q_n\mathbb{Z}$.

- Code parameters:
 - Length: n ,
 - Size: $K := |\mathcal{C}|$;
- Weighted distance [GRS99]: given $a, b \in \mathbb{Z}/q_1\mathbb{Z} \times \cdots \times \mathbb{Z}/q_n\mathbb{Z}$,
 - The *weight* of a is $\mathbf{wt}(a) := \prod_{i:a_i \neq 0} q_i$.
 - The *weighted distance* between a and b is $D(a, b) := \mathbf{wt}(a - b) = \prod_{i:a_i \neq b_i} q_i$.

Chinese Remainder Codes

- $q_1 < q_2 < \dots < q_n$ pairwise coprime;
- $N = \prod_{i=1}^n q_i$ amplitude of the code;
- $K = \prod_{i=1}^k q_i < N$ with $1 \leq k < n$;
- $[C]_{q_i} := C \bmod q_i$



Chinese Remainder code: $\mathcal{CR}(K) = \{([C]_{q_1}, \dots, [C]_{q_n}) : 0 \leq C < K\}.$

Chinese Remainder Theorem: any k correct coordinates uniquely determine C modulo N .

Analogies with Reed-Solomon Codes

Reed-Solomon (MDS)	CR (weighted MDS)
$f \in \mathbb{F}_q[x]_{<k}$	$0 \leq C < K$
Evaluation Codes $\begin{array}{ccc} & / & \backslash \\ [f]_{x-\alpha_1} & \cdots & [f]_{x-\alpha_n} \\ \downarrow & & \downarrow \end{array}$	$\begin{array}{ccc} & / & \backslash \\ [C]_{q_1} & \cdots & [C]_{q_n} \\ \downarrow & & \downarrow \end{array}$
$c = (f(\alpha_1), \dots, f(\alpha_n))$	$c = ([C]_{q_1}, \dots, [C]_{q_n})$
Error weight $\begin{array}{c} e = (e_1, \dots, e_n) \\ \downarrow \text{wt}_H(e) \leq \lfloor \frac{n-k+1}{2} \rfloor \end{array}$	$\begin{array}{c} e = (e_1, \dots, e_n) \\ \downarrow \text{wt}(e) < \sqrt{\frac{N}{K}} \end{array}$
$r = (r_1, \dots, r_n) \leftrightarrow R(x) \in \mathbb{F}_q[x]_{<n}$	$r = (r_1, \dots, r_n) \leftrightarrow R \in \mathbb{Z}/N\mathbb{Z}$
Decoders Welch-Berlekamp	
$f(x) \in \mathbb{F}_q[x]_{<k}$	$0 \leq C \leq K$

Basics on Locally Recoverable Codes

Locality for Codes over the Integers: Formal Definition



Recover an erased coordinate by accessing only a small subset of the remaining coordinates.



Can the notion of locality be generalized to integer codes?

Locality for Codes over the Integers: Formal Definition

? Can the notion of locality be generalized to integer codes?

- $\mathcal{C} \subseteq \mathbb{Z}/q_1\mathbb{Z} \times \cdots \times \mathbb{Z}/q_n\mathbb{Z}$ ~~q_i pairwise coprime~~ integer code,
- $r \in \mathbb{R}^+$.

Coordinate locality

A coordinate $i \in [1, n]$ has *locality* r if there exists a subset $S \subset [1, n] \setminus \{i\}$ such that:

1. (*locality*) $\text{wt}(S) \leq \text{wt}(\{i\})^r = q_i^r$;
2. (*recoverability*) $|\text{Punct}(\mathcal{C}, [1, n] \setminus S)| = |\text{Punct}(\mathcal{C}, [1, n] \setminus (S \cup \{i\}))|$.

- S is a *helper set* for i ,
- $\mathcal{C}$ has locality r if every coordinate has locality r ,
- If $q_1 = \cdots = q_n$, definition specializes to the classical notion of locality.

First (bad) Example: Locality of CR codes

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? *Do there exist (non-trivial) examples of integer-LRCs?*

Given $q_1 < q_2 < \dots < q_n$, consider the CR code $\mathcal{C} := \mathcal{CR}(K)$ with $K = \prod_{i=1}^k q_i$

Locality of CR Codes

Every coordinate of $\mathcal{C}$ has locality

$$r \geq \frac{\log K}{\log q_{\max}}, \quad q_{\max} := \max_i q_i.$$

- A helper set S with weight $< K$ cannot recover a missing coordinate;
- Any set of coordinates of weight $\geq K$ forms an information set.

! Recovering a coordinate is equivalent to recovering the whole codeword.

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Locality of CR Codes

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Constructions of new LRCs: the following constructions preserve locality

- The Cartesian product;
- Concatenation;
- Shortening ;..

Locally Recoverable Integer Codes: Bounds and Constructions

Locally Recoverable Codes over Finite Fields with Hamming Metric

Let $\mathcal{C} \subseteq \mathbb{F}_q^n$ be an $[n, k]$ code with locality r and minimum distance d .

$$\text{LRC bound [GHSY12]: } d \leq n - k - \left\lceil \frac{k}{r} \right\rceil + 2$$

! LRC bound is tight

Tamo-Barg Codes over Finite Fields

- $r \geq 1$ locality parameter;
- $\mathbb{F}_q^* = \cup_i A_i$ with A_i pairwise disjoint and such that $|A_i| = r + 1$;
- $g \in \mathbb{F}_q[x]$ such that $g(y) = m_i$ for all $y \in A_i$
- Given $t \leq \frac{q-1}{r+1}$ such that $k = rt$, let $a = (a_{i,j}, 0 \leq i \leq r-1, 0 \leq j \leq t-1) \in \mathbb{F}_q^k$.

Tamo-Barg codes with locality r

For any $a \in \mathbb{F}_q^k$ define the encoding polynomial of a as $f_a = \sum_{i=0}^{r-1} \sum_{j=0}^{t-1} a_{i,j} g(x)^j x^i$. Then

$$\text{TB}(q-1, k, g) = \left\{ (f_a(\alpha), \alpha \in \mathbb{F}_q^*) \mid a \in \mathbb{F}_q^k \right\}$$

Tamo-Barg codes [TB14]:

- Attain the LRC bound: $d = n - k - \lceil \frac{k}{r} \rceil + 2$;
- Are constructed as a cartesian product of subcodes of Reed-Solomon codes.

LRC bound for Integer Codes

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- $q_1, \dots, q_n$ with $n \geq 2$,
- $N := q_1 \cdots q_n$.

LRC Bound for Integer Codes

Let $\mathcal{C} \subseteq \mathbb{Z}/q_1\mathbb{Z} \times \cdots \times \mathbb{Z}/q_n\mathbb{Z}$ an integer code with cardinality K and locality r . Then,

$$D < N \left(\frac{\max\{q_i\}}{K} \right)^{1+1/r}.$$

LRC bound for Integer Codes

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LRC Bound for Integer Codes

Let $\mathcal{C} \subseteq \mathbb{Z}/q_1\mathbb{Z} \times \cdots \times \mathbb{Z}/q_n\mathbb{Z}$ an integer code with cardinality K and locality r . Then,

$$D < N \left(\frac{\max\{q_i\}}{K} \right)^{1+1/r}.$$

- ❗ The bound yields a real-value, whereas D is a product of subsets of the moduli q_i . Tightening the bound reduces to a multiplicative Subset-Sum problem (NP-hard).

Integer Tamo–Barg Codes

🟡 *Do there exist a family of codes (nearly) meeting the LRC bound?*

Integer Tamo–Barg Codes

❓ Do there exist a family of codes (nearly) meeting the LRC bound?

- $q = (q_1, \dots, q_n)$ and $N := q_1 \cdots q_n$
- Analogue of the evaluation map is the residue map $C \mapsto C \bmod q_i$;
- Analogue of $g(x)$ is an integer G constant modulo products of disjoint subsets of the q_i 's.

Integer Tamo–Barg Codes

Given a s -tuple of integers $A = (A_0, \dots, A_{s-1})$, then for $G \geq 1$ such that $G \bmod q_i$ is a small integer, the *Integer Tamo–Barg code* is:

$$\text{iTB}^A(q, G, D) := \{([C]_{q_1}, \dots, [C]_{q_n}) \mid C = \sum_j a_j G^j \bmod N \text{ with} \\ 0 \leq c < N/D \text{ and } 0 \leq a_j < A_j, \forall 1 \leq j \leq s\}.$$

❗ They arise as subcodes of the Cartesian product of CR codes.

Illustrative Examples integer-TB codes

- $q_1, \dots, q_6$ pairwise coprime, $q = \max_i q_i$, $P_1 = q_1 q_2 q_3$ and $P_2 = q_4 q_5 q_6$, $P := \max Q_i$;
- u_1, u_2 Bézout's coefficients such that $1 = u_1 P_1 + u_2 P_2$;
- For some $\lambda_1, \lambda_2 \geq 1$, $G := u_1 \lambda_1 P_1 + u_2 \lambda_2 P_2$ and $\lambda := \max \lambda_i$

Integer-TB code $\mathcal{C} \subseteq \mathbb{Z}/q_1\mathbb{Z} \times \dots \times \mathbb{Z}/q_6\mathbb{Z}$:

$$\mathcal{C} := \left\{ ([a + bG]_{q_1}, \dots, [a + bG]_{q_6}) \mid 0 \leq a + bG < \frac{N}{D}, 0 \leq a + b\lambda < \frac{N}{Pq} \right\}.$$

Illustrative Examples of integer-TB codes

Integer-TB code:

$$\mathcal{C} := \left\{ ([a + bG]_{q_1}, \dots, [a + bG]_{q_6}) \mid 0 \leq a + bG < \frac{N}{D}, 0 \leq a + b\lambda < \frac{N}{Pq} \right\}.$$

Examples:

	$(q_1, \dots, q_6)$	(λ_1, λ_2)	D	$K = \mathcal{C} $	bound on D
1.	(37, 41, 47, 23, 31, 50)	(3, 1)	$35\,650 = q_4 q_5 q_6$	$\geq 15\,640 \sim 6q_6^2$	$< 556\,474$
2.	(59, 71, 79, 61, 67, 73)	(252, 1)	$298\,351 = q_4 q_5 q_6$	$\geq 21\,252 \sim 4q_6^2$	$< 23\,250\,800$
3.	(97, 107, 113, 101, 103, 109)	(161, 1)	$1\,133\,927 = q_4 q_5 q_6$	$\geq 30\,654 \sim 3q_6^2$	$< 242\,963\,524$

- None of the examples attains the Integer-LRC bound;
- Example 1. is *nearly optimal*;
- If $N \sim P^2$ then $K \geq \frac{1}{u\lambda} \left(\frac{N}{D} - q^{r-1} \right)$.

Open Questions

- ? Derive tighter bounds for Integer-LRCs;
- ? Generalize the construction to an arbitrary number of primes;
- ? Identify new families of (nearly) optimal Integer-LRCs.

Thank you for your attention!

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